THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE UNIT TEST SET B

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. An instrument used to measure weight is a: 1

(a) beam balance only **(b)** barometer
(c) spring balance **(d)** thermometer

2. Charges that do not move by themselves are called: 1

(a) fuses **(b)** static charges **(c)** magnetic poles **(d)** current

3. A land breeze blows: 1

(a) from land to sea at noon **(b)** from land to sea at night
(c) from sea to land at night **(d)** never

4. Lower parts of thunderclouds carry mostly: 1

(a) positive charges **(b)** negative charges **(c)** no charges **(d)** magnetic poles

5. A storm is formed when warm moist air: 1

(a) freezes immediately **(b)** stays still
(c) sinks **(d)** rises, creating low pressure

6. If all particles are removed from a chair, then: 1

(a) nothing of the chair remains **(b)** it weighs less but remains
(c) it turns into a gas **(d)** nothing changes

7. During melting, the interparticle forces of attraction: 1

(a) become stronger **(b)** become weaker **(c)** remain unchanged **(d)** turn infinite

8. Which statement about ice is correct? 1

(a) It is more dense than water
(b) Its particles are farther apart than those in water
(c) It is a gas
(d) It has no particles

9. Bronze is an alloy of: 1

(a) copper and zinc **(b)** lead and zinc **(c)** iron and nickel **(d)** copper and tin

10. Which is a uniform mixture? 1

(a) sand and iron filings **(b)** sea water
(c) oil floating on water **(d)** muddy water

11. A solution of sugar in water is classified as: 1

(a) a pure substance **(b)** a compound
(c) an element **(d)** a homogeneous mixture

12. Which of the following is a solid–liquid solution? 1

(a) soda water **(b)** salt in water **(c)** air **(d)** brass

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. What is muscular force? Give two examples of its use. 2

14. Why are we not crushed by atmospheric pressure? 2

15. What are interparticle spaces? In which state are they minimum? 2

16. Define a compound with an example. 2

17. How does temperature affect the solubility of solids and gases in liquids? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Why do we say ‘weight of the wheat bag is 10 kg’ is not scientifically correct? 3

19. Describe the steps of formation of a storm. 3

20. Why is sodium chloride a compound? State how the evaporation of salt solution gives salt but not sodium and chlorine. 3

21. List three factors that affect how quickly a solute dissolves. 3

22. Explain why the egg floats in salt solution but sinks in plain water. 3

OR

Describe how temperature changes the solubility of sugar in water. 3

23. Explain the origin of friction. 3

OR

A body weighs 50 N on Earth. Find its mass. (1 kg = 10 N) 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

24. Define force. Describe four effects of a force with an example of each. 5

OR

Explain how a spring balance works and how to read its least count, with an example. 5

25. Describe the three states of water with particle diagrams and changes between them. 5

OR

Explain with activities that gases are compressible, but liquids and solids are not. 5

END OF QUESTION PAPER Total: 50 Marks